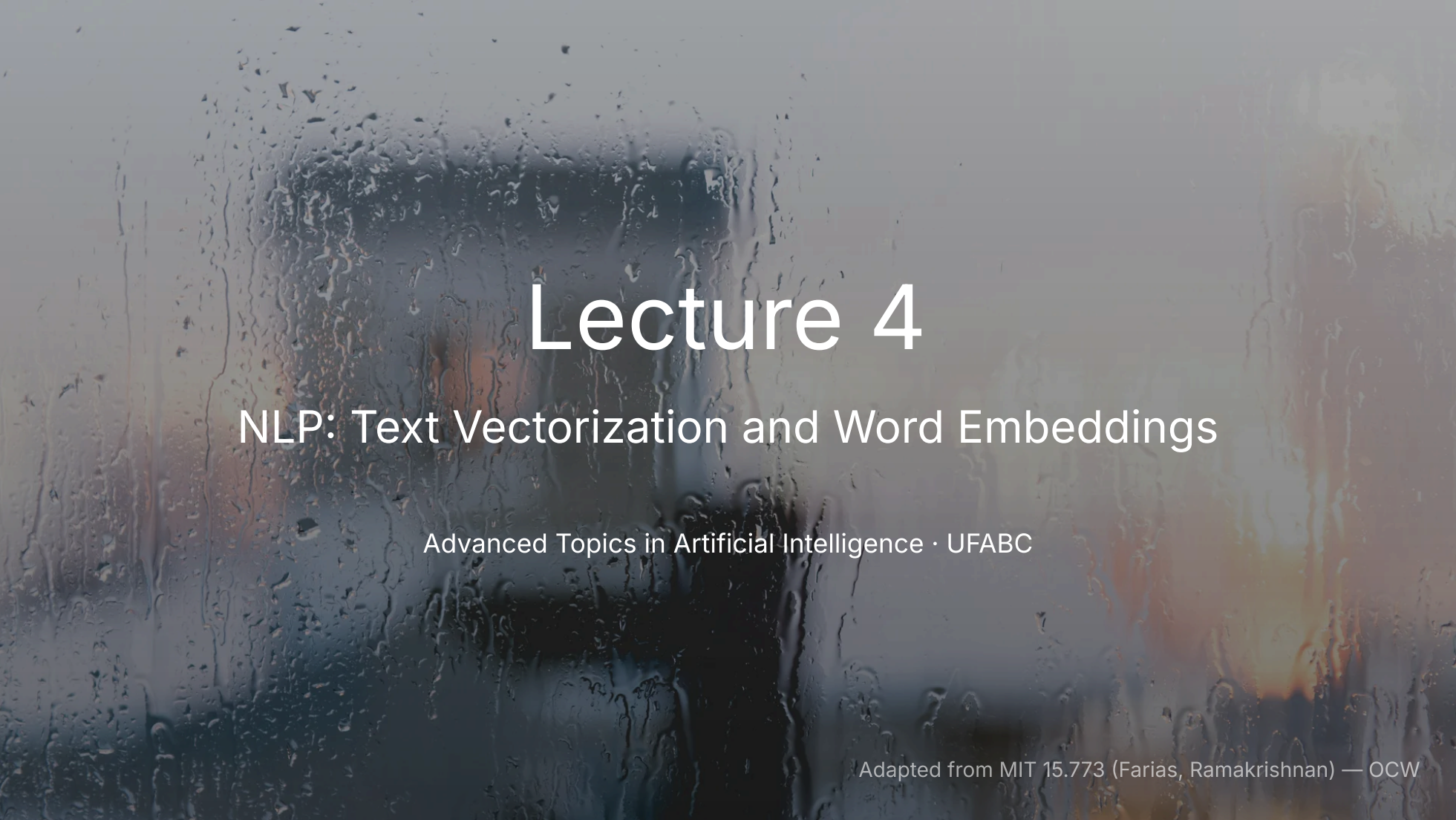


# Lecture 4

## NLP: Text Vectorization and Word Embeddings

Advanced Topics in Artificial Intelligence · UFABC

Adapted from MIT 15.773 (Farias, Ramakrishnan) — OCW

# Part 1 — Why NLP?

# Lecture roadmap

**Part 1 — Why NLP?** Motivation, applications and historical arc

**Part 2 — Text Pre-processing** Pipeline Standardize → Tokenize → Index → Encode

**Part 3 — Bag of Words** Count encoding, multi-hot, limitations

**Part 4 — Word Embeddings** One-hot problems, Word2Vec, GloVe

**Part 5 — Embeddings in Code** `nn.Embedding` (PyTorch), `Embedding` (Keras)

**Part 6 — Application** Text classification with embeddings

# Why NLP?

### Human knowledge lives in natural language

- The Internet is, for the most part, **text**
- Human communication and cultural production are **text**
- Corporate documents, contracts, medical records are **text**

**Imagine** a system that could automatically read and "understand" all of this — extracting patterns, classifying, answering questions, generating content.

### Applications in production

🏷️ **Classification**
Sentiment, intent, routing

📋 **Extraction**
Financials, form data

📝 **Summarization**
Abstracts, bullet points, titles

✍️ **Generation**
Emails, reports, code

💬 **Q&A / RAG**
Chatbots, semantic search

🌐 **Translation**
Multilingual, code-switching

# Historical arc of NLP

📜

**Rules**

up to 1990

- Formal grammars
- Manual syntax parsers
- Performance limited by rule scale

📊

**Statistical / ML**

1990 – 2013

- Bag of Words + SVM
- Hidden Markov Models
- Logistic regression over n-grams

🧠

**Neural Networks**

2014 – 2017

- RNNs and LSTMs
- Word2Vec, GloVe
- Seq2seq with attention

⚡

**Transformers**

2017 – today

- "Attention is All You Need"
- BERT, GPT, T5, LLaMA
- General-purpose LLMs

*"Every time I fire a linguist, the performance of the speech recognizer goes up."* — Frederick Jelinek, IBM

# Big picture: NLP as regression

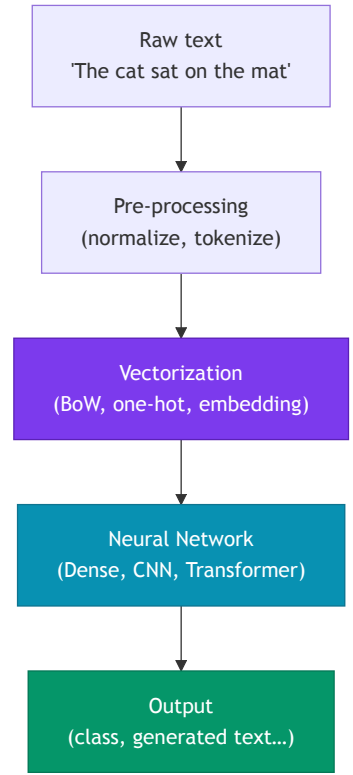
**At its core, it's fancy regression**

$$\hat{y} = f(\mathbf{x}, \theta)$$

- $\mathbf{x}$ = input text (token sequence)
- $\hat{y}$ = text, label, number, ...
- $\theta$ = neural network weights
- $f$ = deep architecture (CNN, RNN, Transformer...)

**Central question of this lecture:**

How do we represent $\mathbf{x}$ (text) as a numerical vector so the network can learn?



# Part 2 — Text Pre-processing

# Pre-processing pipeline


**Standardize**

- Lowercase
- Remove punctuation/accents
- *Stop words* (sometimes)
- Stemming/lemmatization (sometimes)

**Tokenize**

- Split into tokens
- Default: whitespace
- Alternative: subwords (BPE)
- N-grams (bigram, trigram...)

**Index**

- Assign unique integer to each token
- Builds the **vocabulary** $\mathcal{V}$
- Special token `<UNK>` for out-of-vocabulary words

**Encode**

- Map integer → vector
- Simplest: one-hot
- Better: word embedding

# Pre-processing example

**Original text:**

```
"Hola! What do you picture when you think of traveling to Mexico? Sipping a real margarita while soaking up the sun on a laid-back beach in Puerto Vallarta?"
```

**After standardize** (lowercase, remove punctuation, stop words, stemming):

```
`hola picture think travel mexico sip real margarita soak sun laidback beach puerto vallarta`
```

**After tokenize** (by whitespace):

```
`["hola", "picture", "think", "travel", "mexico", "sip", "real", "margarita", "soak", "sun", "laidback", "beach", "puerto", "vallarta"]`
```

**After index** (vocabulary with 50 k tokens):

```
`[4231, 1807, 2943, 8821, 9012, 3344, 771, 18432, 6621, 488, 22301, 993, 19034, 19035]`
```

# One-hot encoding

**Idea:** each vocabulary token gets a vector with 1 at its position and 0 everywhere else.

```
Vocabulary (|V| = 5):  cat, dog, fish, bird, mouse

cat   → [1, 0, 0, 0, 0]
dog   → [0, 1, 0, 0, 0]
fish  → [0, 0, 1, 0, 0]
bird  → [0, 0, 0, 1, 0]
mouse → [0, 0, 0, 0, 1]
```

- Vector dimension = $|\mathcal{V}|$ (vocabulary size)
- Special token `<UNK>` (index 0) for out-of-vocabulary words
- **Sparse** representation — only one non-zero element

**Problem:** for $|\mathcal{V}| = 50{,}000$, each token becomes a 50 k-dimensional vector.

❌ **Very high dimensionality** → many parameters, overfitting

❌ **No semantic similarity** — distance between any two one-hot vectors is always $\sqrt{2}$, regardless of meaning

```python
from sklearn.preprocessing import LabelBinarizer

vocab = ["cat", "dog", "fish", "bird", "mouse"]
lb = LabelBinarizer()
lb.fit(vocab)
print(lb.transform(["cat", "fish"]))
# [[1 0 0 0 0]
#  [0 0 1 0 0]]
```

# Part 3 — Bag of Words

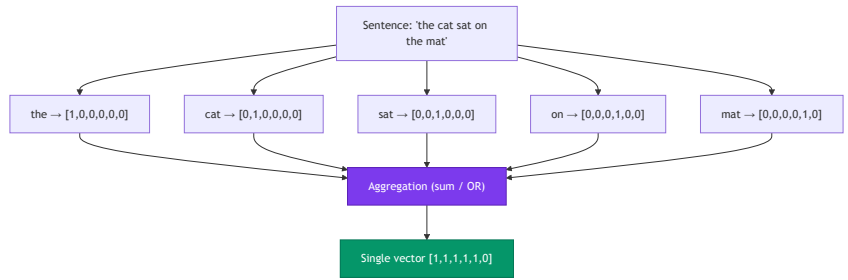
# From per-token to per-document vectors

**The problem:** sentences have different lengths.

A 5-token sentence gives a $5 \times |\mathcal{V}|$ matrix.
A 20-token sentence gives $20 \times |\mathcal{V}|$.

Dense networks need **fixed-size input**.

**Solution:** aggregate (*pool*) all token one-hot vectors into a single vector of size $|\mathcal{V}|$.



# Bag of Words — two variants

**Count encoding** (sum of one-hot vectors)

Counts how many times each token appears.

```
"the cat sat on the mat"
"the mat belonged to the cat"

vocab = [the, cat, sat, on, mat, belonged, to]

Sentence 1: [2, 1, 1, 1, 1, 0, 0]
Sentence 2: [2, 1, 0, 0, 1, 1, 1]
```

If "cat" appears 3×: that component = 3.

**Multi-hot encoding** (OR of one-hot vectors)

Indicates only whether the token is present (0 or 1).

```
"the cat cat cat sat"

Count:     [1, 3, 1, 0, 0, 0, 0]
Multi-hot: [1, 1, 1, 0, 0, 0, 0]
```

Multi-hot ignores frequency; useful when presence matters more than count.

Both are forms of **Bag of Words** — the name comes from treating text as a "bag" of words without order.

# N-grams — capturing local context

**Unigram:** 1 token at a time (standard BoW)

**Bigram:** consecutive pairs of tokens

**Trigram:** consecutive triples of tokens

```
Sentence: "the cat sat on the mat"

Unigrams:  ["the", "cat", "sat", "on", "the", "mat"]

Bigrams:   ["the cat", "cat sat",
            "sat on", "on the", "the mat"]

Trigrams:  ["the cat sat",
            "cat sat on",
            "sat on the",
            "on the mat"]
```

**Why use n-grams?**

✅ Preserves some local context

✅ "not good" ≠ "good" (bigram captures negation)

✅ Simple, works well for text classification

❌ Vocabulary grows explosively: $|\mathcal{V}|^n$

❌ Still ignores long-range dependencies

❌ "good but not great" would need a 4-gram

```python
from sklearn.feature_extraction.text import CountVectorize
cv = CountVectorizer(ngram_range=(1, 2))
X = cv.fit_transform(["the cat sat on the mat"])
```

# Bag of Words limitations

❌ **Loses word order**

"The dog bit the man"
and
"The man bit the dog"

have the **same** BoW vector.

❌ **High dimensionality**

$|\mathcal{V}|$ can be 50 k–500 k.

Every input has that dimension regardless of sentence length.

Many parameters → overfitting, slow.

❌ **No semantics**

"movie" and "film" are equidistant from each other and from "banana".

BoW does not capture that they are related words.

**Summary:** BoW is a useful, simple baseline, but for tasks requiring semantic understanding or long-range dependencies we need something better — **Word Embeddings**.

# Part 4 — Word Embeddings

# The problem with one-hot representations

**Problem 1: High dimensionality**

Vocabulary of 50 k words → 50 k-dimensional vector.
Most elements are zero (**sparse**).

**Problem 2: No semantic distance**

```
"movie"  → [1, 0, 0, 0, 0, ...]
"film"   → [0, 1, 0, 0, 0, ...]
"banana" → [0, 0, 1, 0, 0, ...]

cosine("movie",  "film")   = 0
cosine("movie",  "banana") = 0
# All words are equally "distant"!
```

**The ideal solution**

**Dense** and **compact** vectors (dimension 50–300) where **geometric distance** reflects **semantic distance**.

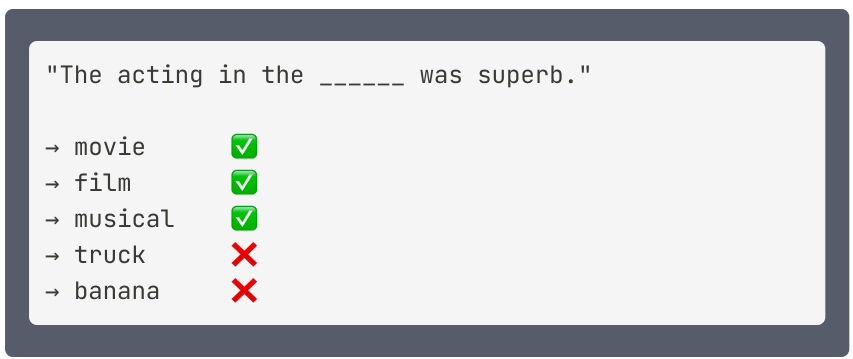
```
"movie"  ≈ [0.2,  0.8, -0.3, ...]  # compact
"film"   ≈ [0.19, 0.79, -0.28, ...] # close to "movie"
"banana" ≈ [-0.6, 0.1,  0.9, ...]  # far from both

cosine("movie", "film")   ≈ 0.97 ✅
cosine("movie", "banana") ≈ 0.12 ✅
```

# The intuition: "you shall know a word by the company it keeps"

*"You shall know a word by the company it keeps."*

*— John Firth, linguist (1957)*

**Similar contexts → related words**




Since "movie", "film" and "musical" appear in the **same contexts**, we infer they are semantically related.

**Key insight**

Count how often two tokens co-occur in the same contexts and learn vectors that **approximate** that co-occurrence matrix.

This is the foundation of two highly influential algorithms:

- **Word2Vec** (Mikolov et al., 2013) — local prediction
- **GloVe** (Pennington et al., 2014) — global co-occurrence

# Word2Vec — two models

**CBOW** (*Continuous Bag of Words*)

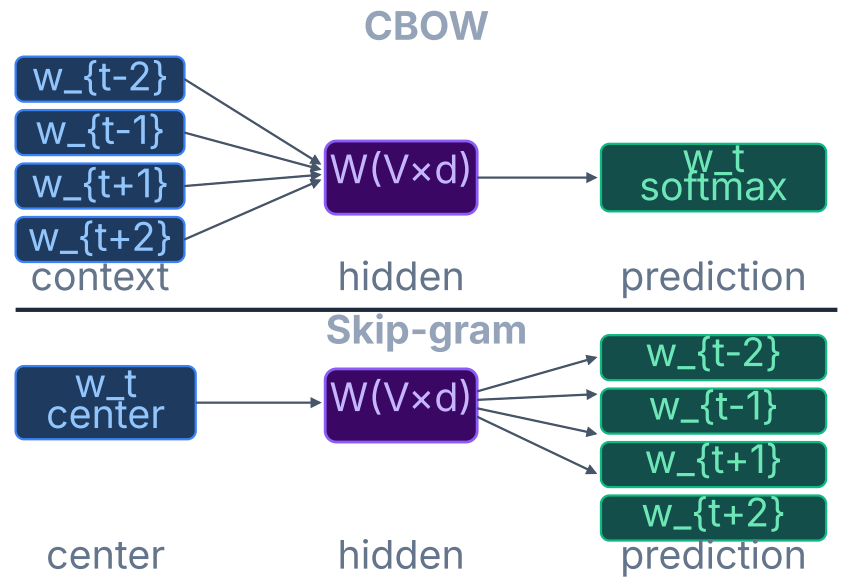
Given the context, predict the center word.

```
Context → Center word
["The", "acting", "was", "superb"]  →  "in the ___"
```

**Skip-gram**

Given the center word, predict the context words.

```
Center word → Context
"movie"  →  ["The", "acting", "was", "superb"]
```




💡 **Negative Sampling** — avoids softmax over |V|: classifies (w, ctx) as real or random using k ≈ 5–20 negatives per step → O(k) instead of O(|V|)

# Word2Vec — Negative Sampling

**Problem:** softmax over 50 k tokens at every step is costly — $O(|\mathcal{V}|)$.

**Solution — Negative Sampling:**
Instead of normalizing over the whole vocabulary, train a **binary classifier**:

$$P(D{=}1 \mid w, c) = \sigma(\mathbf{v}_c^\top \mathbf{v}_w)$$

*"Did this (word, context) pair come from the real corpus?"*

```
Positive pair: ("movie", "acting")  → label 1
Negative pairs (k random samples):
  ("movie", "table")    → label 0
  ("movie", "cloud")    → label 0
  ...
```

**Objective with negative sampling:**

$$\mathcal{L} = \log \sigma(\mathbf{v}_c^\top \mathbf{v}_w) + \sum_{i=1}^{k} \mathbb{E}_{w_i \sim P_n} \left[ \log \sigma(-\mathbf{v}_{w_i}^\top \mathbf{v}_w) \right]$$

✅ Per-step complexity: $O(k)$ with $k \ll |\mathcal{V}|$ (typically $k = 5$–$20$)
✅ Empirically produces embeddings of similar quality to full softmax

More frequent words have a higher probability of being sampled as negatives (distribution $P_n(w) \propto f(w)^{3/4}$).

# GloVe — Global Vectors

**Intuition:** Word2Vec uses local windows. GloVe uses **global co-occurrences** of the corpus.

**Step 1:** Count $X_{ij}$ = how many times word $j$ appears in the context of word $i$ across the entire corpus.

```
          movie  film  banana  ...
movie    [  –,   8432,    12,  ... ]
film     [ 8432,  –,       9,  ... ]
banana   [   12,   9,      –,   ... ]
```

"movie" and "film" co-occur a lot; both rarely co-occur with "banana".

**Step 2:** Learn vectors that approximate the co-occurrence matrix.

**Log-bilinear model:**

$$\log X_{ij} = \mathbf{w}_i^\top \mathbf{w}_j + b_i + b_j$$

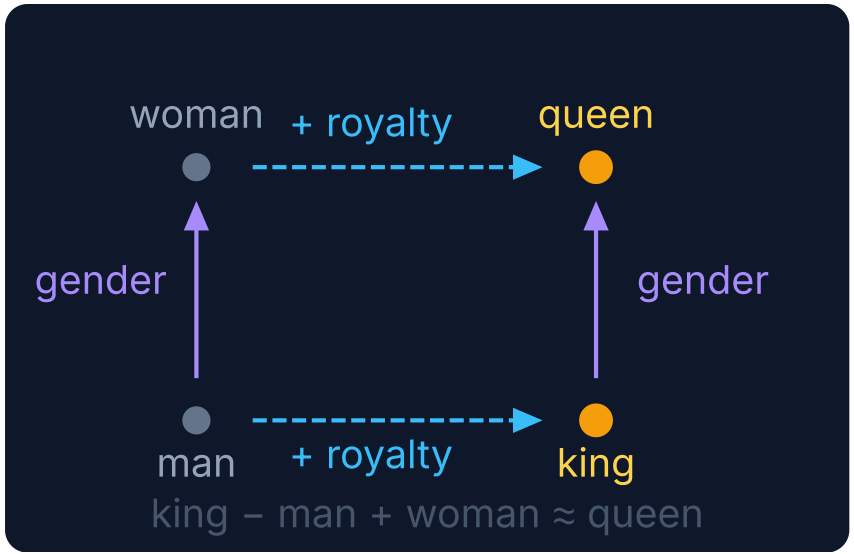
**Objective** (weighted least squares):

$$\mathcal{L} = \sum_{i,j} f(X_{ij}) \left( \mathbf{w}_i^\top \mathbf{w}_j + b_i + b_j - \log X_{ij} \right)^2$$

where $f(X_{ij})$ is a weighting function that prevents very frequent pairs from dominating training.

At the end, biases $b$ are discarded and only $\mathbf{w}$ is used.

# Geometric properties of embeddings




2D projection — each dimension captures a latent concept that emerges from training.

**Types of relations that emerge**

Gender
king → queen · actor → actress · man → woman

Country → Capital
France → Paris · Brazil → Brasília · Japan → Tokyo

Superlative / Degree
good → better → best · big → bigger → biggest

Verb tense
run → ran · sing → sang · go → went

Antonyms
hot ↔ cold · fast ↔ slow · big ↔ small

⚠️ These patterns emerge from the corpus — and also reflect its **biases** (Wikipedia, Common Crawl…).

# Static vs contextual embeddings

**Static embeddings**
(Word2Vec, GloVe, FastText)

- One fixed vector per word, independent of context
- Fast to compute, low memory footprint
- Good for simple classification tasks

❌ "bank" (financial) has the same vector as "bank" (riverbank)

**Contextual embeddings**
(ELMo, BERT, GPT)

- Vector depends on all other tokens in the sentence
- Captures polysemy and long-range dependencies
- Foundation of modern LLMs

✅ "bank" generates different vectors in each context — computed by a **Transformer** (Lecture 6!)

**This lecture** covers static embeddings. **Lecture 5** covers RNNs/LSTMs and **Lecture 6** covers Transformers and contextual embeddings.

# Part 5 — Embeddings in Code

# `nn.Embedding` — PyTorch

**What it is:** a lookup table — maps integer indices to dense vectors.

```python
import torch
import torch.nn as nn

# vocab_size = vocabulary size
# embed_dim  = embedding dimension
embed = nn.Embedding(
    num_embeddings=10_000,  # |V|
    embedding_dim=128        # d
)

# Token indices for a batch of 2 sentences
# (zero-padded to max_len=5)
tokens = torch.tensor([
    [42, 871, 3, 0, 0],
    [7, 1024, 55, 810, 2]
])  # (batch=2, seq_len=5)

out = embed(tokens)  # (2, 5, 128)
print(out.shape)     # torch.Size([2, 5, 128])
```

**Loading pre-trained weights (GloVe)**

```python
import numpy as np

# Assumes glove_matrix: (|V|, d) pre-loaded
embed = nn.Embedding(
    num_embeddings=vocab_size,
    embedding_dim=100
)
embed.weight = nn.Parameter(
    torch.tensor(glove_matrix, dtype=torch.float32)
)

# Optional: freeze weights during fine-tuning
embed.weight.requires_grad = False
```

`nn.Embedding` is equivalent to an `nn.Linear` layer without bias, but with O(1) lookup — it does not perform a full matrix multiplication over all sentences.

# `Embedding` layer — Keras

**Full pipeline with Keras**

```
import tensorflow as tf
from tensorflow import keras

# 1. Vectorization (STI)
vectorizer = keras.layers.TextVectorization(
    max_tokens=10_000,   # |V|
    output_mode='int',   # returns indices
    output_sequence_length=64  # max_len
)
vectorizer.adapt(train_texts)  # learns vocabulary

# 2. Embedding
embed = keras.layers.Embedding(
    input_dim=10_000,    # |V|
    output_dim=128,      # d
    mask_zero=True       # ignore padding
)
```

**Full model**

```
inp  = keras.Input(shape=(1,), dtype='string')
x    = vectorizer(inp)          # (batch, 64)
x    = embed(x)                 # (batch, 64, 128)
x    = keras.layers.GlobalAveragePooling1D()(x)
     # mean over seq_len → (batch, 128)
x    = keras.layers.Dense(64, activation='relu')(x)
out  = keras.layers.Dense(num_classes,
                          activation='softmax')(x)

model = keras.Model(inp, out)
model.compile(
    loss='sparse_categorical_crossentropy',
    optimizer='adam',
    metrics=['accuracy']
)
```

# Global Average Pooling — why it works

After `Embedding`, each sentence is a **matrix (T×d)**. We need a **fixed vector** for the dense layer.

```
"the cat sat"  →  Embedding
[[0.2, -0.1, 0.8, ...],  ← "the"
 [0.7,  0.3, 0.1, ...],  ← "cat"
 [0.4, -0.2, 0.6, ...]]  ← "sat"   shape: (T, d)

GAP → mean over T (dim=1)
→ [0.43, 0.0, 0.5, ...]            shape: (d,)
```

**Why does it work?** GAP is equivalent to a **semantic BoW**: each word contributes its meaning vector, and the average captures the "mean semantic content" of the sentence. Unlike classic BoW (where "movie" and "cinema" are orthogonal vectors), here semantically similar words contribute similar vectors — the result aggregates semantics, not just counts.

**Alternatives to GlobalAveragePooling**

**Flatten** — concatenates all vectors into (T×d), requires fixed length.

**GlobalMaxPooling1D** — maximum per dimension, preserves strongest activations.

**[CLS] token / attention** — Transformer strategies (Lecture 6). BERT uses `[CLS]`.

**Shared limitation:** none of these strategies preserve token **order**. For that, we need RNNs or Transformers.

# Part 6 — Application: Text Classification

# Musical genre classification

**Task:** given a song lyric excerpt, classify it as hip-hop, pop or rock.

```
*"I grew up on the crime side, the New York Times side… Had secondhands, Mom's bounced on old man…"* → hip-hop 🎤
```

```
*"I walked through the door with you But something about it felt like home somehow…"* → pop 🎵
```

**BoW + Embedding architecture**

# Full code — PyTorch

```
import torch, torch.nn as nn, torch.optim as optim

class TextClassifier(nn.Module):
    def __init__(self, vocab_size, embed_dim, num_classes):
        super().__init__()
        self.embed   = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.dropout = nn.Dropout(0.3)
        self.fc1     = nn.Linear(embed_dim, 64)
        self.fc2     = nn.Linear(64, num_classes)

    def forward(self, x):          # x: (batch, seq_len)
        x = self.embed(x)          # (batch, seq_len, embed_dim)
        x = x.mean(dim=1)          # GAP: (batch, embed_dim)
        x = self.dropout(x)
        x = torch.relu(self.fc1(x))
        return self.fc2(x)         # (batch, num_classes)

model     = TextClassifier(vocab_size=10_000, embed_dim=64, num_classes=3)
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=1e-3)

for epoch in range(10):
    for tokens, labels in train_loader:
        logits = model(tokens)
```

# Comparison: BoW vs Embeddings

| | **BoW (count/multi-hot)** | **Embedding + GAP** |
|---|---|---|
| Representation | Sparse ($\|V\|$-dimensional) | Dense (d-dim, $d \ll \|V\|$) |
| Semantics | ❌ No semantic distance | ✅ Similar words are close |
| Word order | ❌ Ignored | ❌ Ignored (with GAP) |
| Parameters | None (engineered) | $\|V\| \times d$ (learned) |
| Transfer | Only with external embeddings | ✅ Pre-train on large corpus |
| Cost | Low | Moderate |

**To capture word order and long-range dependencies** → sequential architectures: RNNs and LSTMs **(Lecture 5)**, and ultimately Transformers **(Lecture 6)**.

# Summary

**Pre-processing**

- S→T→I→E pipeline
- Standardization, tokenization, vocabulary
- One-hot encoding: sparse, no semantics

**Bag of Words**

- Count and multi-hot encoding
- N-grams for local context
- Limitations: order lost, high dimensionality

**Word Embeddings**

- Word2Vec (skip-gram, negative sampling)
- GloVe (global co-occurrence, least squares)
- Dense vectors with semantic distance

**In code**

- `nn.Embedding` (PyTorch)
- `Embedding` + `TextVectorization` (Keras)
- GlobalAveragePooling1D as weighted BoW

**Remaining limitations**

- No order with GAP
- Static embeddings: polysemy
- → We need Transformers!

**Lecture 5 — RNNs and LSTMs**

- Hidden state and sequential processing
- BPTT and the vanishing gradient problem
- LSTMs and GRUs as solutions

# Next lecture

**Lecture 5 — RNNs and LSTMs**

- Hidden state and sequential processing
- BPTT and the vanishing gradient problem
- LSTM: memory cells and gates
- GRU: more efficient variant
- Applications: classification and seq2seq

**For this week**

- Notebook `lec04-embeddings.ipynb`:
  - Pre-processing pipeline with Keras/PyTorch
  - Train embeddings from scratch vs pre-trained GloVe
  - Musical genre classification
  - Embedding visualization with t-SNE

# Thank you! Questions?

CCM-109 · Deep Learning · UFABC

Adapted from MIT 15.773 Hands-on Deep Learning (Farias & Ramakrishnan, 2024)